Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December-2023/January-2024
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Explain briefly about Scripts.	[2]
b)	Write a short note on Associative Arrays.	[3]
c)	What is a Directory?	[2]
d)	List out File Permission commands.	[3]
e)	What is a Signal?	[2]
f)	Explain briefly about a reliable signal.	[3]
g)	What is a Pipe?	[2]
h)	Discuss briefly about “pclose” library function.	[3]
i)	Define Sockets.	[2]
j)	Explain briefly about “fcntl” system call.	[3]

PART – B

(50 Marks)

2.a) List out and briefly write about Shell Responsibilities.
b) Explain in detail about String handling and Mathematical Functions available in Linux. [5+5]

OR

3.a) Write a detailed note on Security by File Permissions.
b) Discuss in detail about Process Utilities and Disk Utilities. [5+5]

4. Explain the system calls used for file I/O operations with syntax and illustrative example. [10]

OR

5.a) What is a File? Explain in detail the Structure of a File system.
b) Explain with examples for creating, removing and changing Directories. [5+5]

6.	Demonstrate fork( ) and wait( ) system calls with example codes.	[10]
	OR	
7.a)	Discuss in detail about system( ) system call.	
b)	What is a Signal Function? Explain about “kill” and “sleep” Functions.	[4+6]
8.	Explain in detail Kernel support for Message Queues in Linux.	[10]
	OR	
9.	Explain the client/server example implementation using FIFOs.	[10]
10.	Explain in detail about APIs for Shared Memory.	[10]
	OR	
11.	Write a detailed note on Socket system calls for Connection Oriented Protocol.	[10]

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